

Practical Exercises: The National Information Ecosystem

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Module purpose

Participants learn to see official statistics as a national information ecosystem rather than as the output of one national statistical office. They identify the institutions, data assets, technologies, flows, users, rules, capabilities and risks that determine whether information is fit for public decisions. They then use that map to locate realistic AI opportunities and the dependencies that must be addressed first.

Performance outcome

Given a country scenario or an approved national domain, the participant can produce a scoped ecosystem map with stable identifiers, evidence references, critical flows, user needs, quality and governance risks, coordination gaps and a justified statement of where AI may or may not add value.

Module	A1
Competencies	C1 Statistical purpose and public value; C3 stewardship; C4 quality; C6 governance; C8 communication.
Entry level	No prior AI expertise required; basic familiarity with a statistical organization or sector.
Exit level	L1 guided practitioner, with an optional L2 workplace assignment.
Structured time	6 hours, delivered as one day or two 3-hour sessions.
Practice share	Approximately 65 percent.
Primary evidence	Completed ecosystem map, flow analysis, risk statement and oral defence.
Phase 1 link	Uses the concepts and record types of Phase 1D.

Package contents

Part	Use
Use	Learner activity pack. Complete the exercises using the supplied Kintaba evidence before consulting any decision guide.

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Fictional country case Republic of Kintaba

Kintaba is a lower-middle-income country with 24 million people. Agriculture employs a large share of the rural population and is exposed to drought, pests and price volatility. The government needs earlier district crop-production estimates for food-security planning. The following case is fictional and contains deliberate gaps and contradictions.

Actor ID	Actor	Reported role and situation
ACT-01	National Statistics Directorate	Runs the agricultural census every ten years and an annual production survey. Its frame is based on the last census.
ACT-02	Ministry agriculture statistics unit	Publishes seasonal crop estimates from extension reports and a farmer sample. Methods differ by region.
ACT-03	Extension service	Collects monthly crop-condition forms from local officers. Coverage is better near roads.
ACT-04	Land administration agency	Maintains registered parcels, but customary holdings are often absent.
ACT-05	Meteorological service	Operates 82 stations; 19 had substantial missing observations last season.
ACT-06	National food-security secretariat	Combines crop, market, nutrition and rainfall information for cabinet alerts.
ACT-07	Agricultural research institute	Runs crop-variety trials at six research stations and selected farmer fields.
ACT-08	University geospatial laboratory	Produces experimental crop maps from free satellite imagery with external project funding.
ACT-09	Market information service	Collects weekly prices by phone from selected markets.
ACT-10	Mobile network operator	Holds aggregated mobility and mobile-money indicators; no standing data-sharing agreement exists.
ACT-11	Farmer organizations	Report that official estimates arrive late and rarely include small irrigated plots or gender-disaggregated access information.
ACT-12	Government digital service	Hosts ministry systems but does not manage sector metadata or statistical methods.

Case evidence

Evidence ID	Extract
EVD-01	Statistical law assigns national coordination to ACT-01 and allows designated sector producers.
EVD-02	A memorandum requires ACT-02 to send survey microdata and metadata to ACT-01 within 90 days; last year only tables were sent.
EVD-03	ACT-02 reports 4.8 million tonnes of maize; ACT-01 reports 4.2 million for the same season. Definitions and revision status are not shown.
EVD-04	ACT-08 reports 86 percent crop-classification accuracy on 600 labelled fields in three accessible districts.
EVD-05	ACT-07 trial results show two varieties with higher mean yields under controlled water management. Adoption conditions outside trial sites are not assessed.
EVD-06	Food-security alerts are due six weeks before the annual survey estimate is released.
EVD-07	No common identifier links holdings, parcels, extension reports and survey units.
EVD-08	The geospatial project has eight months of funding remaining and one trained analyst.

Exercise 1 Map actors authority and evidence

Time: 45 minutes. Work in groups. Use ACT and EVD identifiers. Produce an actor register and a short authority analysis.

Task	Required response
A	Assign each actor its relevant roles: source owner, producer, methodological authority, steward, coordinator, technology provider, funder, oversight or user. Multiple roles are allowed.
B	For each assignment, label it E, R or U and cite an evidence ID or the case statement.
C	Identify three authority or accountability questions that must be resolved before a national crop-estimation integration project.
D	Name one actor missing from the case and explain why it matters.

Submission template

Actor ID	Role	Evidence label and reference	Authority or coordination issue

Exercise 2 Inventory and assess assets

Time: 45 minutes. Identify at least eight assets, covering four different source types. Do not confuse the producing institution with the asset or the published output.

Asset ID	Asset	Type	Owner	Main quality or fitness issue
AST-01	Annual agricultural production survey	Survey	ACT-01	Frame age and release timing.

Challenge questions

- Which asset is most authoritative for each quantity, and why?
- Which assets are timely but potentially biased or incomplete?
- Which research or geospatial result is being generalized beyond its evidence?
- Which metadata or reference asset is missing?
- Which assets can be combined now, only conditionally, or not yet?

Exercise 3 Trace the crop-estimate flow

Time: 60 minutes. Trace the flow from farm conditions to a cabinet food-security decision. Include at least five actors, three assets and four transformations or handoffs.

Step	Source actor and asset	Transformation or handoff	Control and owner	Failure or delay	Destination
1					
2					
3					
4					
5					
6					

Required conclusion

Identify the two most consequential failure points. For each, state the affected user decision, consequence, present evidence, missing evidence, responsible role and a realistic non-AI improvement.

Exercise 4 User need risk and AI opportunity

Time: 40 minutes. Screen the proposal: “Use generative AI and satellite imagery to produce district crop estimates automatically.” Do not redesign the entire national system. Produce a bounded recommendation.

Screen	Question
Purpose	What exact decision and statistical product are being improved?
Baseline	What current method, timing, cost and error evidence will the proposal be compared with?
Data	Which sources are permitted, representative, sufficiently documented and stable?
Quality	What benchmark, districts, crops, subgroups, seasons and tolerances are required?
Authority	Who owns the estimate, method, review and release decision?
Human oversight	Which qualified person can challenge or override the result?
Security and confidentiality	What information may enter each tool and environment?
Sustainability	What skills, funding, infrastructure, supplier and fallback dependencies exist?
Decision	Proceed to scoping, investigate dependencies first, or do not pursue? Under what conditions?

Individual assessment prompt

In no more than 500 words, define the ecosystem boundary for improving the timeliness of Kintaba's district crop-production estimates. Identify the critical actors, assets and flow; distinguish evidenced, reported and unknown claims; state the leading fitness-for-use problem; and recommend one next action. Your answer must explain why that action comes before or instead of an AI pilot.